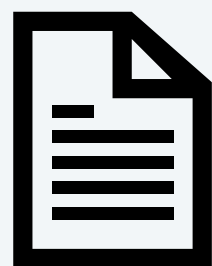


A) Source Data

MIMIC-CXR

Image + Report

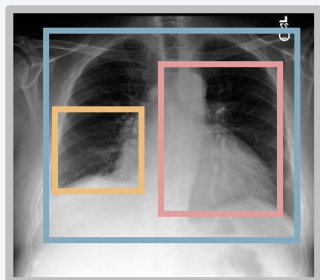


Full Image

Report

GEMeX-ThinkVG

Grounded VQA + Reasoning Traces



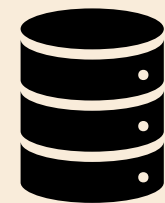
Full Image

Q: Is there evidence...?
A: Yes
Grounded Trace: ...

Q: What does the...?
A: Fluid accumulation.
Grounded Trace: ...

Q: How does the...?
A: Increased
Grounded Trace: ...

B) Alignment and Reformulation



Study-level
Alignment

Reformulation



From Question-level Traces
to Study-level Evidence

Initial Consolidated Evidence

Study ID: p157xxx



The size of
the cardiac
silhouette...

Bounding Box: [43, 153, 154, 227]
Regional Observation: Small right pleural effusion at the right lung base and...

Bounding Box: [116, 143, 287, 245]
Regional Observation: Mild enlargement of the cardiac silhouette, suggesting...

Bounding Box: [43, 35, 320, 263]
Regional Observation: Mild diffuse bilateral interstitial pulmonary opacities...

C) Multi-Stage Checking



Self-Checking



Cross Verification



Expert Adjudication



Accuracy



Completeness



Visual-Text
Consistency

D) Aggregation and Assembly

Anatomy-Guided Aggregation

Image Crop and Data Assembly

MMRad-22K

Multimodal Evidence



Lung Evidence



The size of the
cardiac silhou-
ette has minim-
ally increased...

Mediastinal Evidence



The cardiac silhou-
ette shows m-
ild enlargement,
indicating...



The size of
the cardiac
silhouette...